

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



Department of Information and Communication Technology

LAB Report

Lab Report No: 01

Lab Report Title: Abstract Classes vs Interfaces: Design and Multiple Inheritance.

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By-

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Submitted To-

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Date of Performance:

Date of Submission:

Lab Title : Abstract Classes and Interfaces in Java .

Objective :

To study abstract classes and interfaces in Java .

Theory :

Abstract classes and interfaces are used to achieve abstraction in Java . An abstract class can have abstract and non-abstract methods . An interface supports multiple inheritance by allowing a class to implement more than one interface .

Program Code :

```
abstract class Shape {  
    abstract void draw();  
}  
  
interface Color {  
    void fill();  
}  
  
class Circle extends Shape implements Color {  
    public void draw() {  
        System.out.println("Drawing Circle");  
    }  
    public void fill() {  
        System.out.println("Filling Color");  
    }  
    public static void main(String[] args) {  
        Circle c = new Circle();  
        c.draw();  
        c.fill();  
    }  
}
```

Output :

Drawing Circle
Filling Color

Conclusion :

Abstract classes and interfaces provide abstraction in Java. Interfaces support multiple inheritance while abstract classes do not.

- Abstract classes are used for related objects that share DNA and logic. These define identity ("is-a") and lets you share actual code.
- Interface is used for unrelated objects that just share a skill (e.g. filling). It defines capability ("can-do") and acts as a strict contract.